

Satellite Communications

ECE 4644 / 5610 Spring 2026

Week #15

April 27 – May 1, 2026

Instructor: J. Michael Ruohoniemi

Topics

- Timeline to end of semester
- Notes on Course Project presentation
- VSATs - continued
- On Board Processing (OBP)
- VSAT system design / Example
- Propagation effects

Timeline to end of semester (1/2)

- Apr 27 (M): Lecture, tutorial, *Off-campus students - call for proctoring arrangement*
- Apr 29 (W): Guest lecture - Prof. Tim Pratt (VT), HW#9 due, HW#10-Demo issued
- May 1 (F): *Select day (M or W) next week to make CP presentation* > email instructor

Timeline to end of semester (2/2)

- May 4 (M): CP presentations, Lecture, CP Final Report due, HW#9 returned, Tutorial – HW Review
- May 6 (W): Last class – CP presentations, Course review, *Off-campus students: Information on proctoring arrangement due*
- May 11 (Mon): Final exam, 7:45 – 9:45 AM

Course Project Presentation (1/4)

- Each team member should participate, in-person or virtually
- A Powerpoint presentation is preferred
- Email the instructor your slide stack by 2:00 PM on the day of presentation
- But also bring your presentation on a USB drive

Course Project Presentation (2/4)

- For the final exam all students will be responsible for knowing the 'main idea' of each presentation
- Slide stacks will be posted to the Canvas website for review

Course Project Presentation (3/4)

- Presentations may be made on Monday, May 4 or Wednesday, May 6 (last class)
- Email the instructor with your preference by 5 PM on Friday, May 1
- Length of the presentation: 10-12 min
Including 1-2 minutes for questions
- Terms could be modified

Course Project Presentation (4/4)

Presentation do's and don'ts:

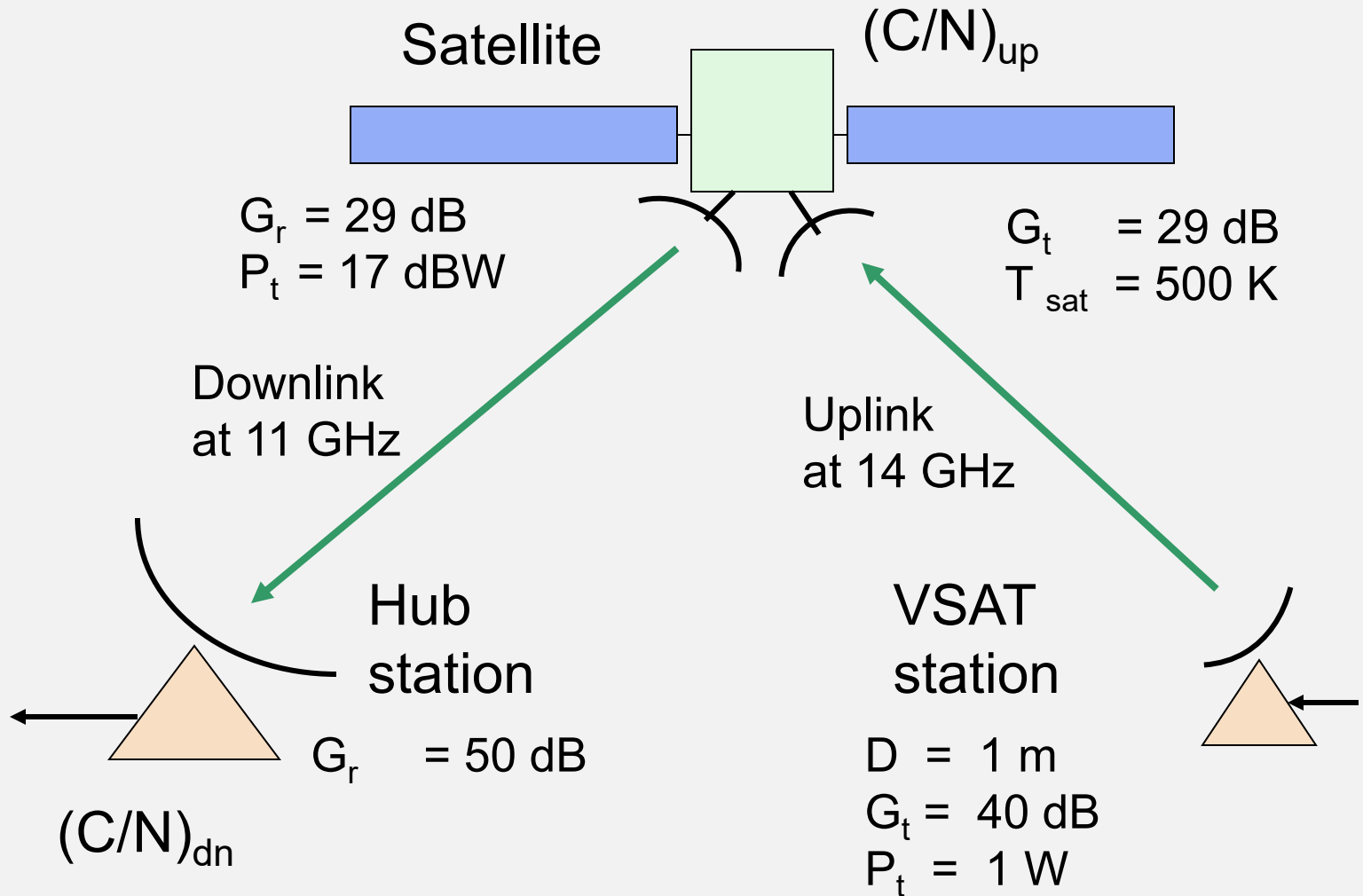
- Do limit your slides to average no more than one per minute and Don't go over the time limit
- Don't try to cover all the details that were given in your report; Do provide an informative overview; Do select one particularly interesting aspect of your system for more discussion
- Do provide a summary that says something about the future of the system
- Do practice your presentation

VSATs - Overview

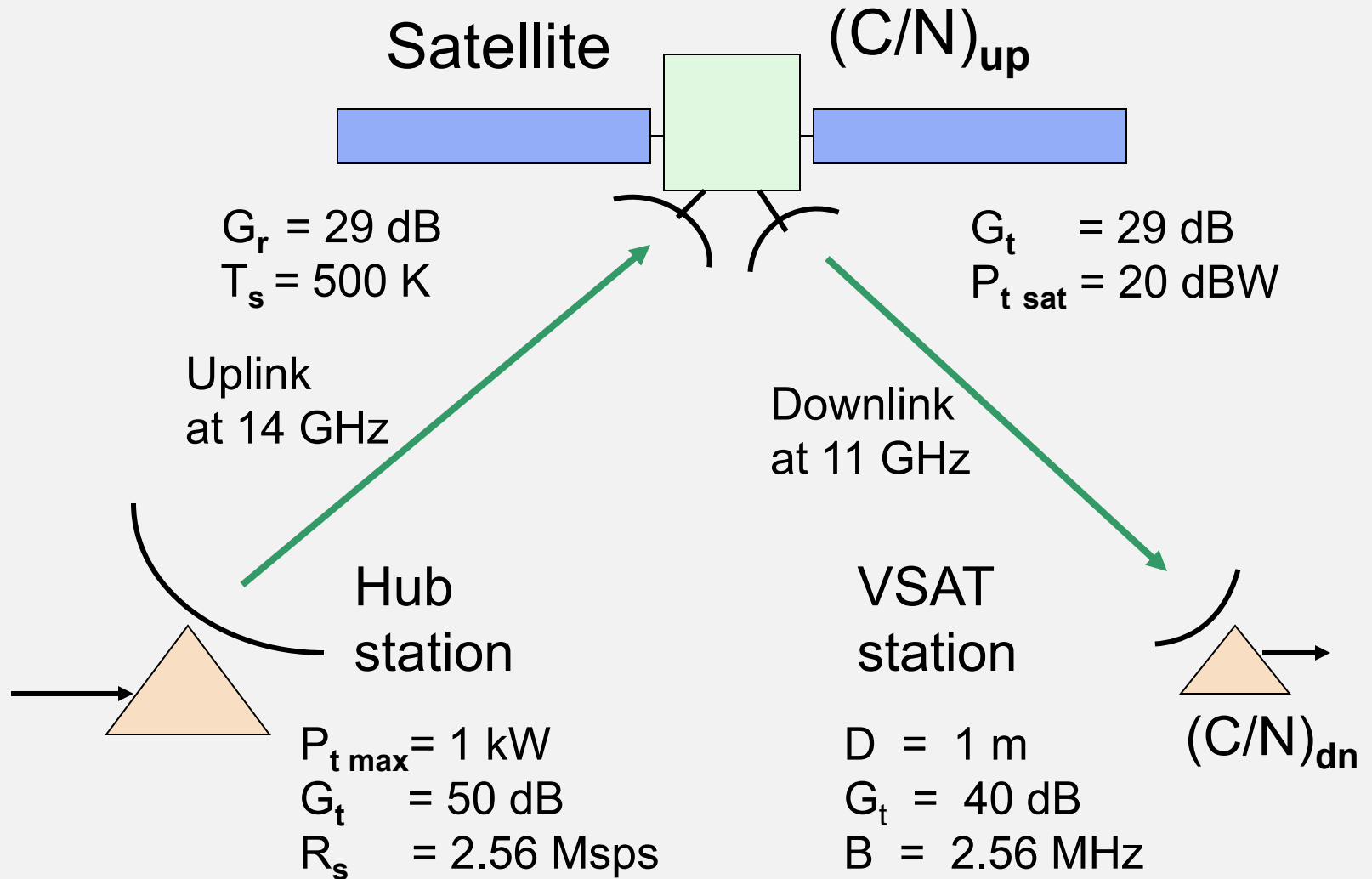
- Very Small Aperture Antenna
- 1 – 2 m in diameter and operates on a few watts
- With GEO satellites operated in a star network (with a central hub) or mesh network
- VSAT systems are used to link businesses to perform point of sale transactions
- NGSO systems operate with smaller phased array antennas

VSAT Multiple Access Schemes

- With Star mode it is more usual to operate with
 - FDMA from the VSAT on the inbound
 - TDM from the hub on the outbound
- Reminder that we are taking the hub as the reference for describing inbound / outbound
- The VSAT transmits on the inbound and receives on the outbound



Inbound Link: VSAT to Hub – FDMA at 128 ksps



Outbound Link: Hub to VSAT – TDM at 2.56 Msps

Star Network with FDMA-TDM: Inbound

- Note that each route (inbound or outbound) consists of an uplink and downlink
- The inbound is processed via FDMA with each VSAT assigned a transponder channel
- Hub receives many, perhaps hundreds, of downlink VSAT channels

Star Network with FDMA-TDM: Outbound

- For the outbound, the hub assembles the individual VSAT message packets into a single, wideband TDM stream
- Each VSAT receives the stream and peels off its message packets
- Remember the distinction between Time Division Multiplexing (TDM) and TDMA
 - TDM is not a multiple access technique

Star Network with FDMA-TDM: Notes

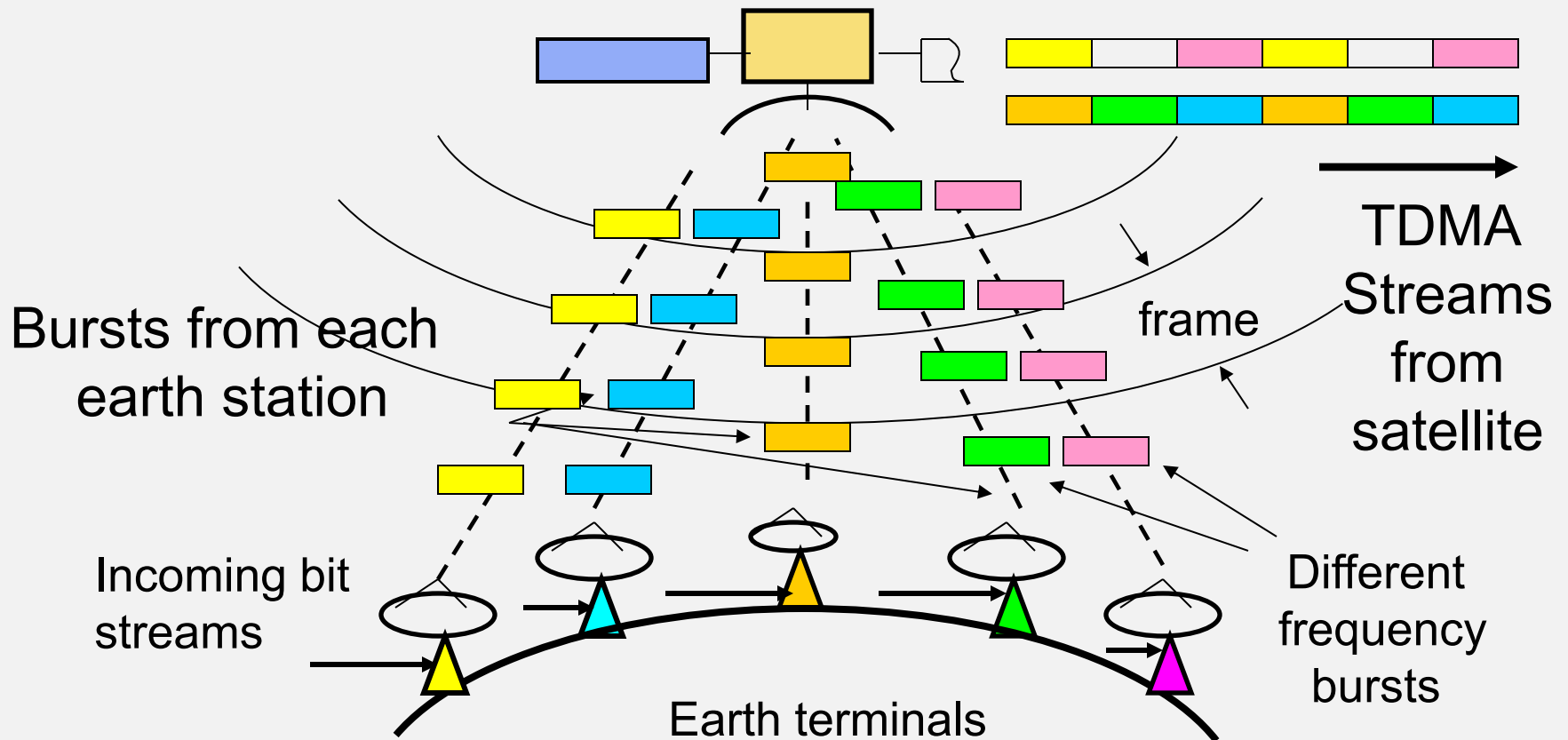
- A single transponder might carry both the inbound narrowband FDMA and the outbound wideband TDM
- Use of FDMA with many channels on the inbound necessitates output standoff on the transponder (3-7 dB)
- Because of cost considerations, the system is likely to operate near margin and suffer occasional outages

Star Network with MF-TDMA

- Popular with Ka band (30/20 GHz) systems
- Reduce FDMA on the inbound by bundling sets of VSAT signals into TDMA
- So the hybrid system on the inbound is MF-TDMA
- More bandwidth efficient but requires higher VSAT transmit power, more complexity
- The sets of VSAT signals are still handled by FDMA; outbound from hub is still TDM

MF – TDMA Internet Access

2 FDMA
channels



On-Board Processing (OBP)

- With On-Board Processing (OBP) received signal is demodulated to baseband at the satellite
- Then processed, reassembled, and transmitted onwards
- Compare with bent pipe transponder
- OBP is flexible, in that uplink and downlink are effectively independent
- Adds complexity (and smartness)

Mesh Network: On-Board Processing (OBP)

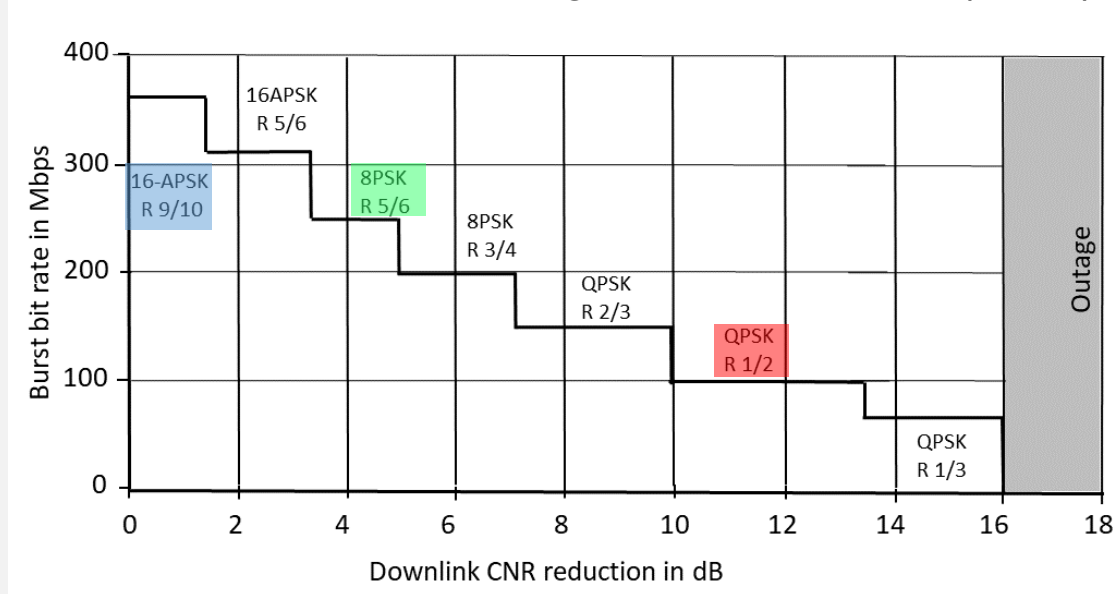
- Can be used to collect FDMA signals from many uplinks:
 - Receive each (narrow-band) uplink signal
 - Process each uplink signal
 - Remove errors, Read addresses
 - Repackage signals into a wideband TDM stream
 - Transmit TDM stream

Mesh Network: On-Board Processing (OBP)

- Advantageous, especially for Ka and V band systems with their sensitivity to rain
- With OBP additional FEC and/or a reduction in modulation order can be applied just to those links affected by rain
- Called *Adaptive Coding and Modulation* (ACM)
- Throughput will decrease on those links
- This is better than having outages

Fig 11.4 page 598 class text

Example of Adaptive Coding and Modulation (ACM)



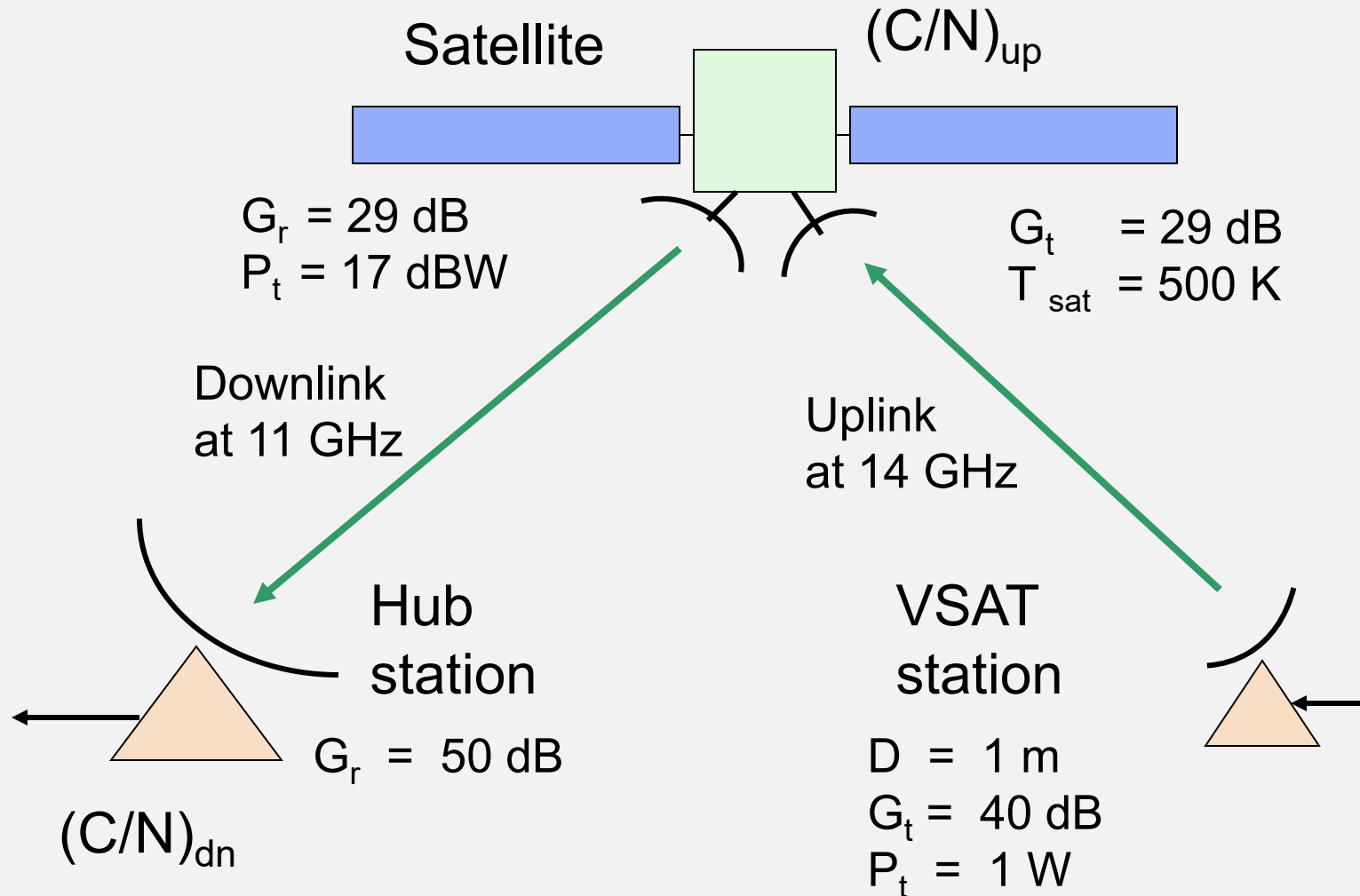
Example of Adaptive Modulation and Coding to combat rain attenuation on downlink

OBP Advantages

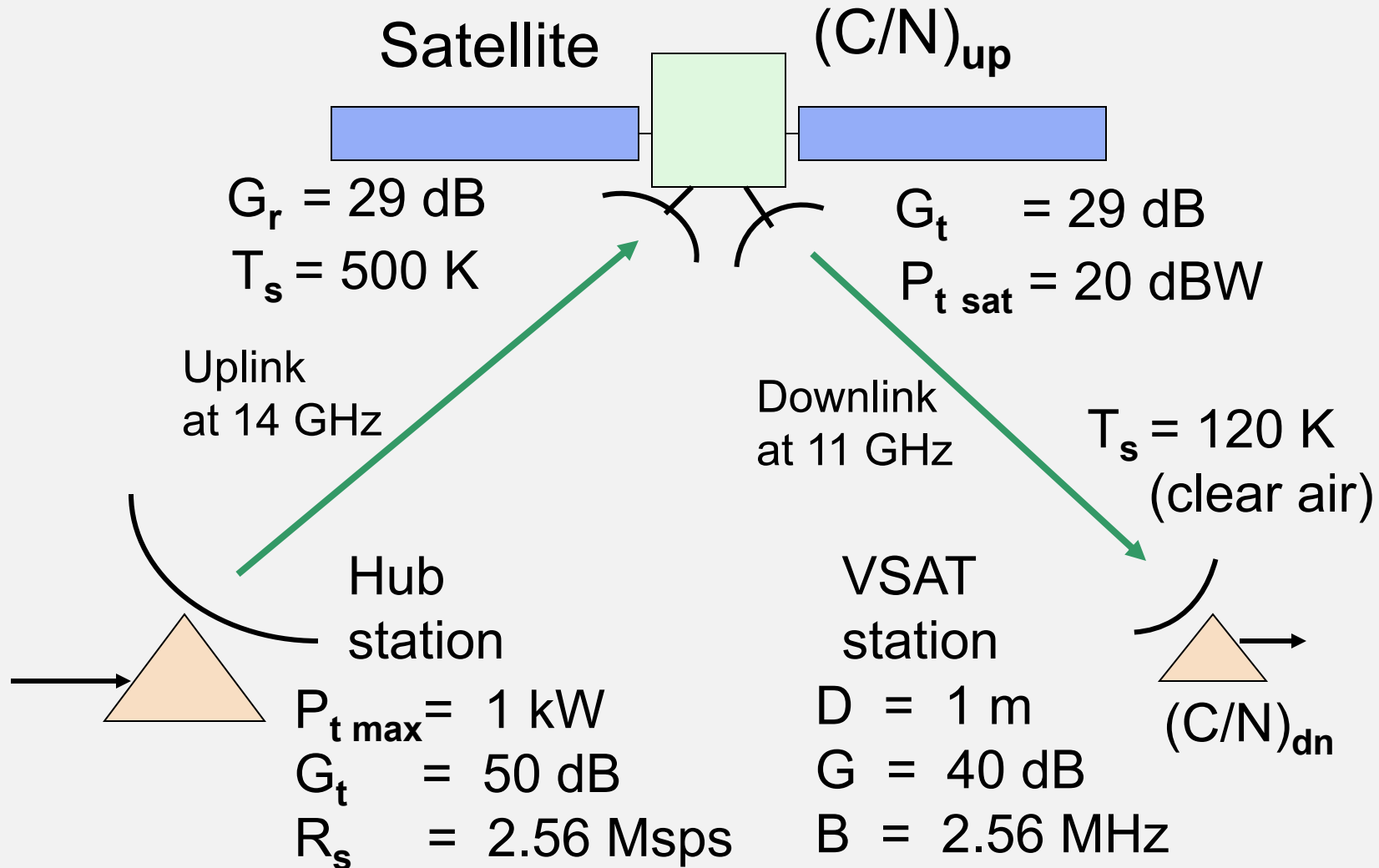
- Processing at satellite reduces number of hops
- Detection of signal at satellite enables
 - U/L power control to be exercised
 - On-board routing of traffic (especially with multiple beams)
 - Error detection / correction of uplink signals
- Adaptive coding and modulation can compensate for reduced CNR due to rain

VSAT System Design Example

- Requirements:
 - Design a VSAT Star network using 1 m terminals with 1 W transmitters
 - Transmit and receive 128 kbps data streams in both directions
- Two links must be designed:
 - Inbound link from VSAT to Hub
 - Outbound link from Hub to VSAT



Inbound Link: VSAT to Hub – FDMA at 128 ksps



Outbound Link: Hub to VSAT – TDM at 2.56 Msps

VSAT System Design Considerations

- There are four RF links in a two-way VSAT system
- Inbound:
 - uplink from VSAT to satellite transponder
 - downlink from transponder to hub
- Outbound:
 - uplink from hub to satellite transponder
 - downlink from transponder to VSAT

VSAT System Design Considerations

- Each link has its own C/N ratio and requires a separate link budget
- Uplink and downlink C/N ratio are then combined to give an overall $(C/N)_o$
- One for the hub (inbound), one for the VSAT (outbound)
- Bit errors ratios can then be calculated and checked

VSAT System Design - Outbound Link

- Large hub station transmits single TDM bit stream
- TDM stream is continuous sequence of packets with addresses
- Each VSAT recovers its own packets
- Specification: $\text{BER} \leq 10^{-6}$ with 4 dB decrease in C/N due to rain
- Link margin is small – outages will occur

Outbound Link

- We must assume some TDM bit rate to begin design
- Let's assume we have active 40 VSAT stations
- Outbound message bit rate is $40 \times 128 \text{ kbps} = 5.12 \text{ Mbps}$
- Assume no FEC to start

Outbound Link

- Then transmitted bit rate = 5.12 Mbps
- Use QPSK
- Then symbol rate = 2.56 Msps
- Noise bandwidth of VSAT receiver = 2.56 MHz
- Assume $(C/N)_{up}$ is much larger than $(C/N)_{dn}$
- Then $(C/N)_o \approx (C/N)_{dn}$

Outbound Link

- Weak link is the downlink from satellite to VSAT station
- Analyze this link for noise bandwidth $B_n = 2.56 \text{ MHz}$
- Use RRC filters with $\alpha = 0.2$
- Single QPSK symbol stream at 2.56 Msps
- $B_{\text{occ}} = R_s (1+\alpha) = 2.56 \times 1.2 = 3.072 \text{ MHz}$

Outbound Downlink (Satellite – VSAT)

- Use 3 dB back off in transponder

$$P_{\text{out}} = 20 - 3 = 17 \text{ dBW}$$

- Transponder output is 17 dBW = 50 W
- Take 1/5 of this power = 10 W = 10 dBW for outbound downlink
- Build in 5 dB of margin to cover miscellaneous losses and 4 dB loss due to rain (worst case?)

Outbound Downlink Budget (Satellite - VSAT)

- Power budget

P_t satellite	10 dBW
G_t on axis	29 dBW
Edge of satellite beam loss	- 3 dB
Path loss at 11 GHz	- 206 dB
G_r VSAT	40 dB
Link margin	<u>- 5 dB</u>
P_r VSAT	-135 dBW

Outbound Downlink Noise Budget

- Downlink noise budget (VSAT)

k	- 228.6 dBW/K/Hz
T_s (clear air) 120 K	20.8 dBK
B_n 2.56 MHz	<u>64.1 dBHz</u>
N	-143.7 dBW

Outbound Downlink C/N

- $(C/N)_{dn} = -135.0 \text{ dBW} - -143.7 \text{ dBW} = 8.7 \text{ dB}$
- This value includes decrease of 4 dB in $(C/N)_{dn}$ for rain in downlink (recall 5 dB link margin)
- So in clear air we are better by 4 dB and $(C/N)_{dn} = 8.7 \text{ dB} + 4 \text{ dB} = 12.7 \text{ dB}$

Outbound Downlink Analysis

- With the possibility of rain causing 4 dB of loss in downlink $(C/N)_{dn} = 8.7$ dB
- May be too low for comfort
- Switching to $\frac{1}{2}$ rate FEC improves effective C/N by 6 dB (coding gain)
- Then worst case effective $(C/N)_{dn}$
$$(C/N)_{dn} = 8.7 + 6 = 14.7 \text{ dB}$$
- Better than clear air $(C/N)_{dn} = 12.7$ dB but throughput has been halved

Outbound Link Analysis

- Overall $C/N = (C/N)_{dn}$ if $(C/N)_{up} > 30$ dB
- Allow 0.5 dB implementation margin
- Then worst case effective $(C/N)_{dn}$
$$(C/N)_{dn} = 14.7 - 0.5 = 14.2 \text{ dB}$$
- Find BER for QPSK signal for this C/N
- $BER = Q(5.13) = 10^{-7}$ in worst case
- Surpasses minimum $BER = 10^{-6}$

Outbound Downlink Analysis

- Check clear air BER (98% of time)
- Apply same 6 dB of coding gain and 0.5 dB of implementation margin
- $(C/N)_o \text{ effective} = 12.7 + 6 - 0.5 = 18.2 \text{ dB}$
- $BER = Q(8.13) = 0$
- There are no errors in clear air
- All digital links are designed to give zero errors in normal operation

Outbound Downlink Analysis

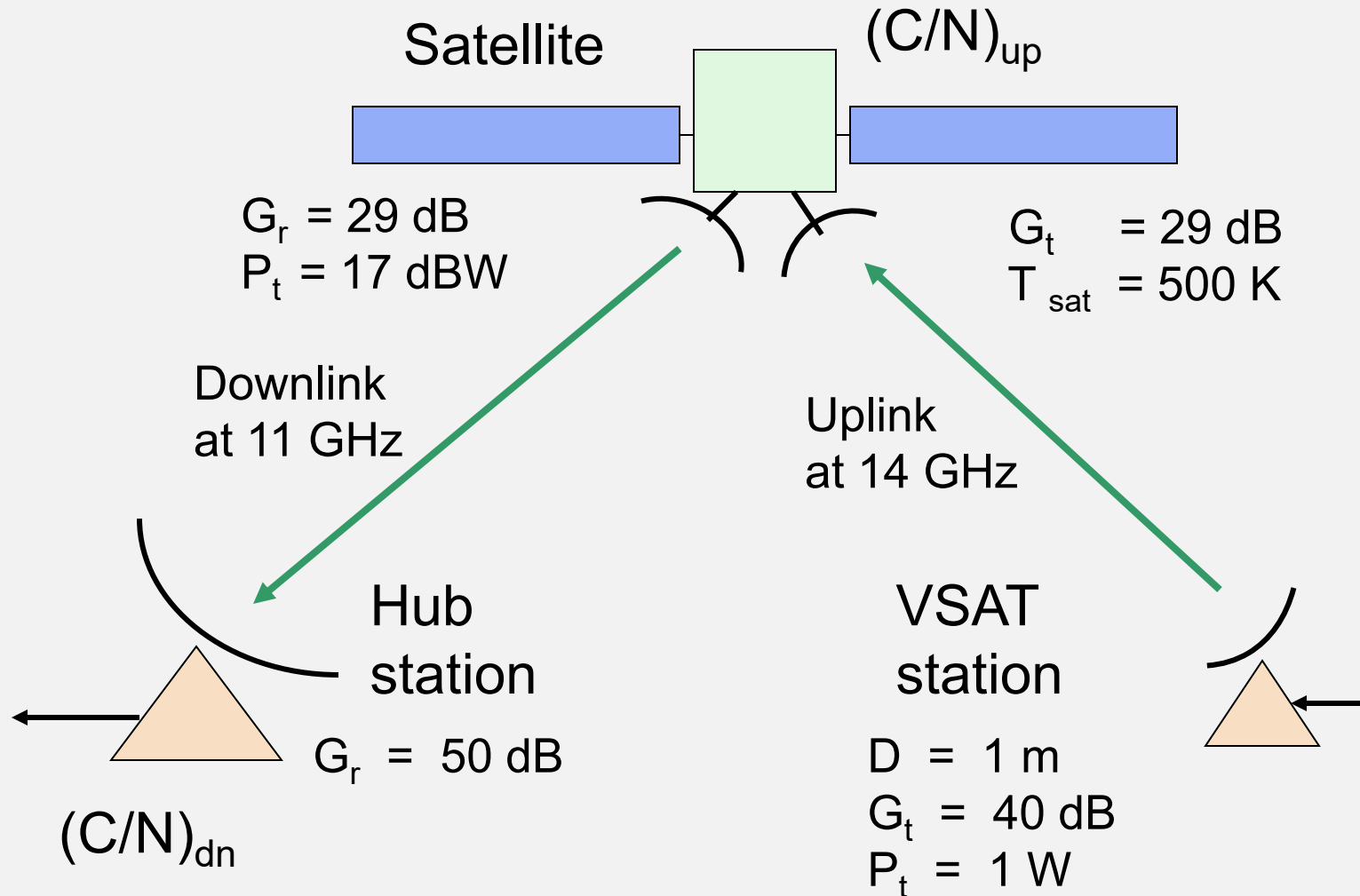
- We are operating acceptably in both clear air $(C/N)_o = 18.2$ dB and heavy rain $(C/N)_o = 14.2$ dB with $\frac{1}{2}$ rate FEC
- Suggests that in clear air we could decrease the FEC rate and increase throughput
- Or switch to a higher modulation scheme, e.g. BPSK to QPSK, QPSK to OPSK
- Or some combination of the two
- That is, apply ACM

Inbound Link – VSAT to Hub

- We assume 40 active VSAT stations
- Inbound data bit rate is 128 kbps
- Use FDMA – best for small stations
- Use RRC filters with $\alpha = 0.2$
- Use QPSK and half rate FEC coding
- Coding gain of ≥ 6 dB is possible

Inbound Link – VSAT to Hub

- Transmitted bit rate = 256 kbps
- Symbol rate = 128 ksps
- Noise bandwidth of hub receiver = 128 kHz
- Assume $(C/N)_{dn}$ is much larger than $(C/N)_{up}$
- Then $(C/N)_o \approx (C/N)_{up}$
- Hub has large, high gain antenna



Inbound Link: VSAT to Hub – FDMA at 128 ksps

Inbound Uplink (VSAT – Satellite)

- Weak link is from VSAT station to satellite
- Analyze this link for noise bandwidth $B_n = 128$ kHz
- Occupied bandwidth is

$$B_{occ} = 128 \times 1.2 = 153.6 \text{ kHz}$$

- Allow 10 kHz guard band between carriers
- 40 FDMA carriers occupy bandwidth

$$40 \times (153.6 + 10) \text{ kHz} = 6.544 \text{ MHz}$$

Inbound Downlink (Satellite – Hub)

- Use 3 dB back off in transponder

$$P_{\text{out}} = 17 \text{ dBW} = 50 \text{ W}$$

- Take 2/5 of this power for inbound downlink = 20 W
- Each FDMA carrier is allocated $20 \text{ W} / 40 = 0.5 \text{ W}$ at satellite output
- Allow 5 dB margin to cover all losses
- Rain may cause 4 dB reduction in C/N

Inbound Uplink Budget (VSAT to Satellite)

- Uplink power budget

P_t VSAT (1 W)	0 dBW
G_t on axis	40 dB
Path loss at 14 GHz	- 207 dB
G_r satellite	29 dB
Edge of satellite beam loss	- 3 dB
Link margin	<u>- 5 dB</u>
P_r VSAT	-146 dBW

Inbound Uplink Noise Budget

- Uplink noise budget (Satellite)

k	-228.6 dBW/K/Hz
T_s (clear air) 500 K	27.0 dBK
B_n 128 kHz	<u>51.1 dBHz</u>
N	-150.5 dBW

- $(C/N)_{up} = -146.0 \text{ dBW} - -150.5 \text{ dBW}$
 $= 4.5 \text{ dB}$

Inbound Uplink Analysis

- With coding gain of 6 dB for $\frac{1}{2}$ rate FEC and implementation margin of 0.5 dB we can boost this to

$$(C/N)_{\text{eff}} = 10.0 \text{ dB}$$

- $\text{BER} = Q(3.3) = 10^{-4}$
- This design does not meet spec ($\text{BER} \leq 10^{-6}$)
- Recall that this analysis included 4 dB of rain attenuation

Inbound Uplink Analysis

- The BER is too high when rain affects the uplink
- We need $C/N_{\text{eff}} = 13.6 \text{ dB}$ for $\text{BER} = 10^{-6}$
- We must find $\sim 3.6 \text{ dB}$ of additional $(C/N)_{\text{up}}$
- Options:
 - Increase EIRP of VSAT station
 - Increase satellite antenna size
 - Decrease bit rate (and noise power)

Inbound Uplink Analysis

- Or combine options to gain 3.6 dB, e.g.,
split evenly between power and antenna gain
 - Increase P_t to + 1.8 dBW = 1.5 W
 - Increase antenna diameter by 0.9 dB to 1.2 m

Inbound Uplink Analysis

- Worst case $(C/N)_{\text{eff}} = 13.6 \text{ dB}$
 - BER for QPSK = 10^{-6}
- In clear air, $(C/N)_{\text{eff}} = 13.6 + 4.0 = 17.6 \text{ dB}$
 - BER for QPSK = $Q(7.59) = 1.5 \times 10^{-14}$
- There are no errors in clear air
- As expected

Inbound Downlink Analysis

- We should check downlink to hub station
- Each FDMA carrier is allocated 0.5 W at satellite output
- Allow 5 dB margin to cover all losses
- Includes rain attenuation on uplink or downlink = 4 dB
- Calculate link budget and noise budget for downlink

Inbound Downlink Budget (Satellite to Hub)

- Power budget

P_t per VSAT channel	- 3.0 dBW
G_t on axis	29 dB
Edge of satellite beam loss	- 3 dB
Path loss at 11 GHz	- 206 dB
G_r Hub	50 dB
Link margin	<u>- 5 dB</u>
P_r Hub	-138 dBW

Inbound Downlink Noise Budget

- Downlink noise budget (Hub)

$$k \quad -228.6 \text{ dBW/K/Hz}$$

$$T_s \text{ (clear air) } 120 \text{ K} \quad 20.8 \text{ dBK}$$

$$B_n \quad 128 \text{ kHz} \quad \underline{51.1 \text{ dBHz}}$$

$$N \quad -156.7 \text{ dBW}$$

- $(C/N)_{dn} = -138.0 \text{ dBW} - -156.7 \text{ dBW}$
 $= 18.7 \text{ dB} < \text{much higher than } (C/N)_{up}$

Overall Link Analysis

- We have obtained acceptable effective (C/N) and BER on all four links individually
- We should check on $(C/N)_o$ when there is attenuation
- Usually assume one of uplink / downlink is in clear air while the other has 4 dB of attenuation
- Determine percentage of time each link has $BER > 10^{-6}$, report on outage time

Inbound Link Analysis

- Increase in VSAT power (to 3 W) may be needed to ensure $\text{BER} = 10^{-6}$ on inbound link
- We have met all system requirements
- Note that transponders are not used efficiently
- We used 1/5 of outbound transponder power in 3.1 MHz bandwidth (TDM stream)
- We used 2/5 of inbound transponder power in 6.544 MHz bandwidth (FDMA channels)

VSAT System Design Summary

- Set parameter initially to suitable values
- Use single TDM stream outbound, FDMA inbound
- Calculate link budgets and noise budgets for outbound and inbound links

VSAT System Design Summary

- Adjust parameters until requirements are met for worst case
- Check operation in clear air conditions:

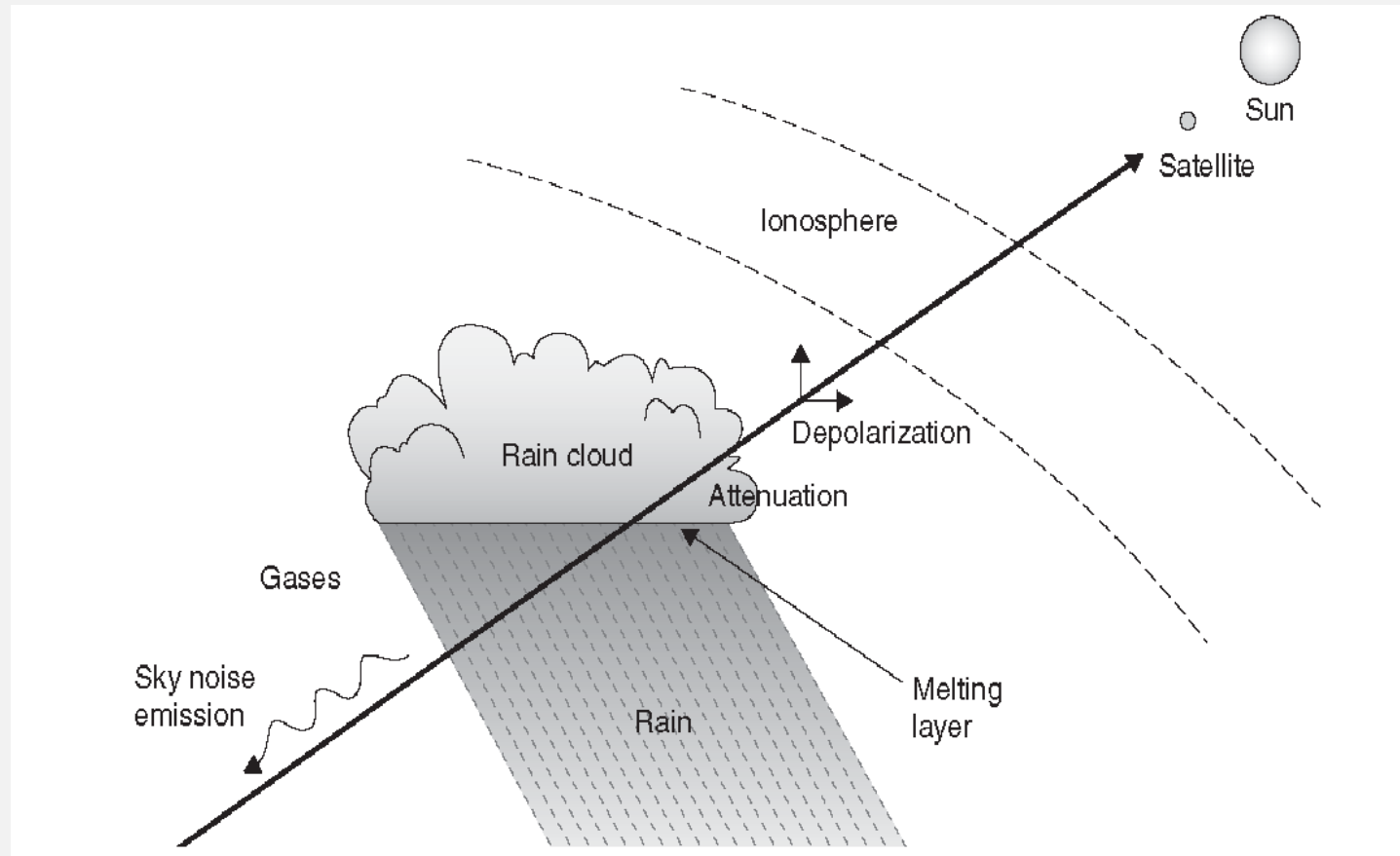
$$\text{BER} \Rightarrow 0$$

- Estimate outage times due to rain attenuation for terminals in different geographical locations

Propagation Effects

- Two propagation effects:
 - (1) Clear air
 - Absorption by oxygen and water vapor
 - Attenuation is small below 50 GHz
 - (2) Rain
 - Will cause outages above ~ 9 GHz
 - Need a way to predict how often link will fail

Sources of Loss in the Atmosphere

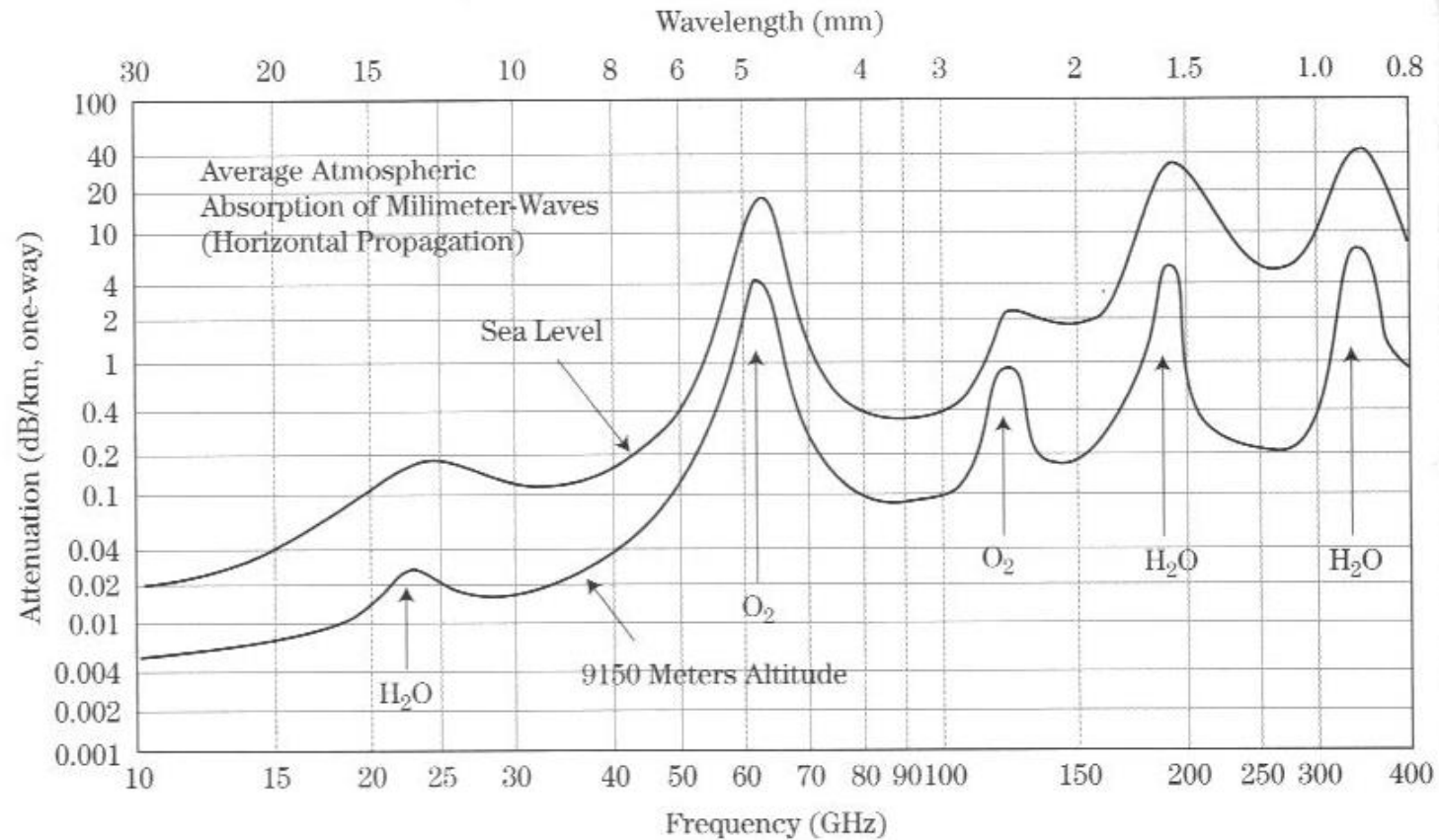


Taken from Figure 7.3 of Pratt and Allnut, 3rd ed.

Propagation in the Atmosphere

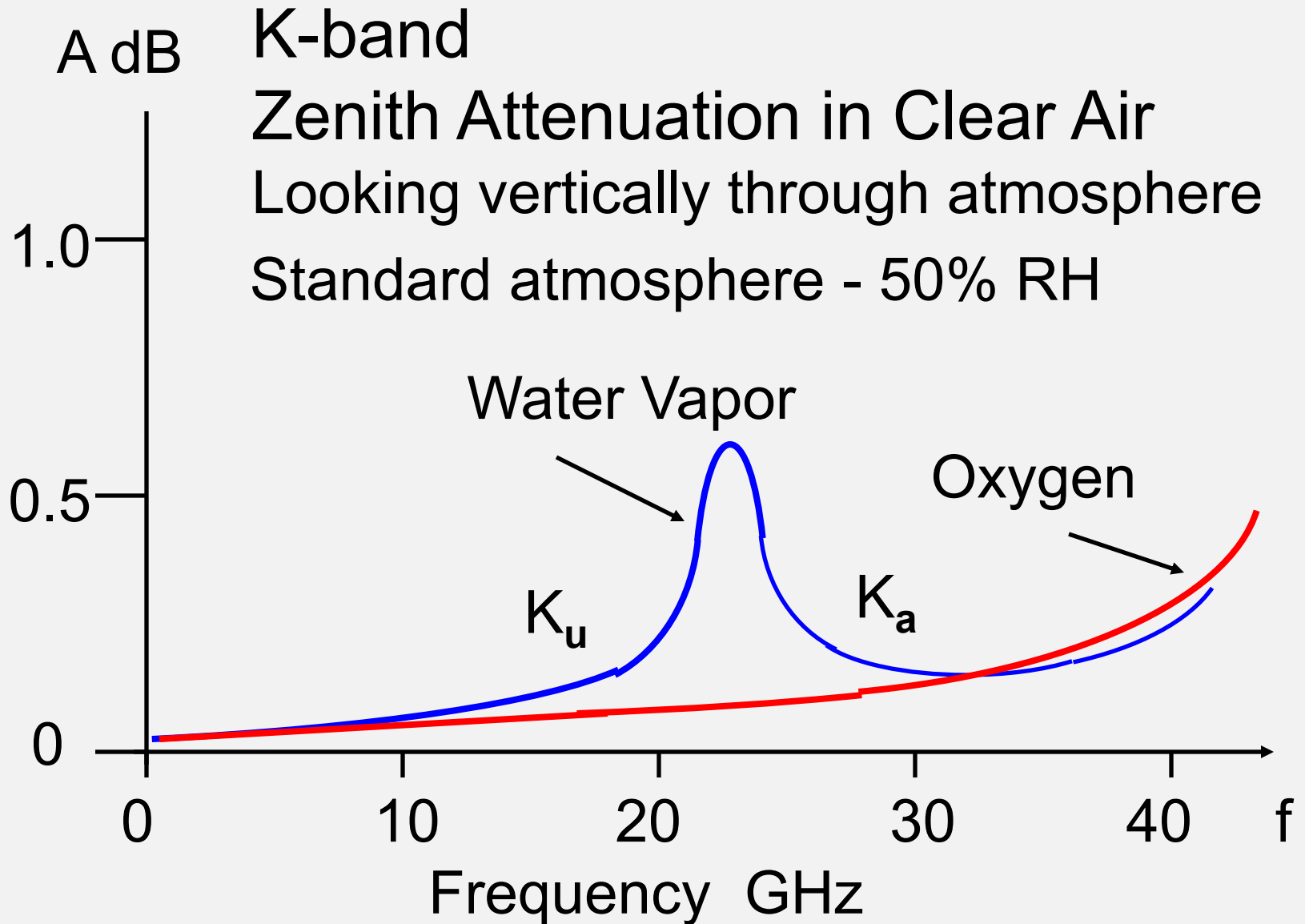
- The atmosphere constitutes a propagation medium with a range of effects on signal: absorption, scattering, refraction, depolarization
- Loss of signal energy from the intended path of propagation makes up the atmospheric loss term, L_{atm} , in the link equation
- Generally atmospheric effects increase with frequency

Atmospheric Attenuation: Variation with Frequency



Propagation in Clear Air

- Attenuation bands are due to resonant absorption of radio wave with oxygen and water vapor molecules
- Attenuation depends on relative humidity
- In damp conditions attenuation at 20 GHz may exceed attenuation at 30 GHz
- Attenuation increases as elevation angle is reduced – more atmosphere in slant path



Zenith Attenuation in Clear Air

Looking vertically through atmosphere

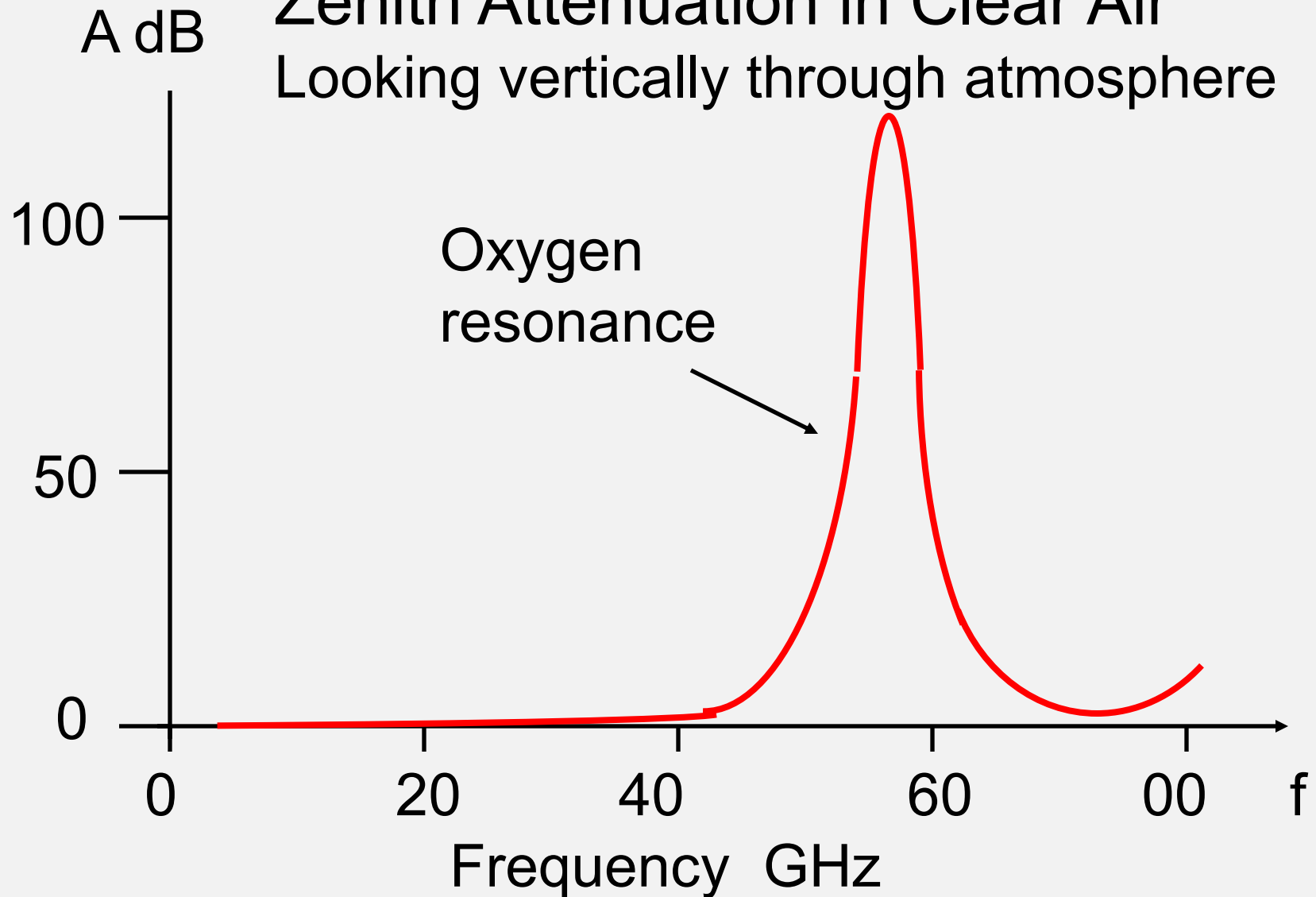


Illustration of Satellite Constellation using Inter-Satellite Links (ISLs)



Credit: Kratos

Propagation in Clear Air

- Oxygen molecules resonate between 55 GHz and 60 GHz
- Very high attenuation
- Frequency band can be used for inter-satellite communication outside Earth's atmosphere
- Water vapor resonance is minor by comparison

Elevation Scaling Law

- Attenuation will depend on the path length through the atmosphere
- Reducing elevation angle increases path length and attenuation
- Elevation angle scaling law in dB ($EL > 10^\circ$):

$$A(EL_1) / A(EL_2) = \operatorname{cosec} EL_1 / \operatorname{cosec} EL_2$$

(cosecant is the inverse of sine)

Elevation Scaling Law

- Example:

Zenith attenuation at 22 GHz is 0.6 dB

Slant path attenuation with EL = 30° is

$$A(30^\circ) = A(90^\circ) \times \operatorname{cosec} 30^\circ / \operatorname{cosec} 90^\circ$$

$$\begin{aligned} A(30^\circ) &= 0.6 \text{ dB} \times \sin 90^\circ / \sin 30^\circ \\ &= 1.2 \text{ dB} \end{aligned}$$

